

MAS 812: Nonparametric Methods

Lecture 4: Linear Rank Statistics and Asymptotic Theory

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Learning Objectives

By the end of this lecture, you will be able to:

- Define linear rank statistics and understand their structure
- Identify common score functions and their properties
- State and apply the Hájek representation theorem
- Understand the asymptotic normality of rank tests
- Derive the asymptotic distribution of common rank statistics
- Apply these results to hypothesis testing

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1 Introduction to Linear Rank Statistics

Linear rank statistics form the foundation of most nonparametric tests. They provide a unified framework for understanding and deriving the asymptotic properties of rank-based procedures.

1.1 Motivation

Recall the Wilcoxon rank-sum statistic:

$$W = \sum_{i=1}^m R_i$$

where R_i are the ranks of the first sample in the combined sample. This is a special case of a more general class of statistics.

Definition 1.1 (Linear Rank Statistic). Let $X_1, \dots, X_m$ and $Y_1, \dots, Y_n$ be two independent samples, and let $R_1, \dots, R_m$ be the ranks of the X 's in the combined sample of size $N = m + n$. A **linear rank statistic** has the form:

$$S = \sum_{i=1}^m a_N(R_i)$$

where $a_N(1), \dots, a_N(N)$ are **scores** (constants generated by a score function).

More generally, for a single sample with ranks $R_1, \dots, R_n$:

$$S = \sum_{i=1}^n c_i a_N(R_i)$$

where c_i are regression constants.

1.2 Examples of Linear Rank Statistics

- **Wilcoxon rank-sum:** $a_N(i) = i$ (linear scores)
- **Mann-Whitney form:** Equivalent to $a_N(i) = i - \frac{N+1}{2}$
- **Median test:** $a_N(i) = \text{sign}(i - \frac{N+1}{2})$
- **Van der Waerden (normal scores):** $a_N(i) = \Phi^{-1}\left(\frac{i}{N+1}\right)$
- **Exponential scores (Savage):** $a_N(i) = \sum_{j=N-i+1}^N \frac{1}{j} - 1$

2 Score Functions

Definition 2.1 (Score Function). A **score function** $\varphi : (0, 1) \rightarrow \mathbb{R}$ is a measurable function that generates the scores via:

$$a_N(i) = \varphi\left(\frac{i}{N+1}\right) \quad \text{or} \quad a_N(i) = \mathbb{E}[\varphi(U_{(i)})]$$

where $U_{(i)}$ is the i -th order statistic from a $\text{uniform}(0, 1)$ distribution.

2.1 Types of Score Functions

1. **Linear scores:** $\varphi(u) = u$ (Wilcoxon)
2. **Quadratic scores:** $\varphi(u) = (u - 1/2)^2$ (for scale alternatives)
3. **Normal scores:** $\varphi(u) = \Phi^{-1}(u)$ (Van der Waerden)
4. **Sign scores:** $\varphi(u) = \text{sign}(u - 1/2)$ (median test)
5. **Exponential scores:** $\varphi(u) = -\ln(1 - u) - 1$ (Savage)
6. **Wilcoxon signed-rank:** $\varphi(u) = u - 1/2$ (one-sample)

Example 2.1 (Score Functions for Common Tests).

Test	Score Function	Use
Wilcoxon rank-sum	$\varphi(u) = u$	Location shift
Van der Waerden	$\varphi(u) = \Phi^{-1}(u)$	Normal scores
Savage	$\varphi(u) = -\ln(1 - u) - 1$	Exponential distributions
Mood	$\varphi(u) = (u - 1/2)^2$	Scale alternatives
Median test	$\varphi(u) = \text{sign}(u - 1/2)$	Robust location

2.2 Properties of Score Functions

For asymptotic theory, we typically require:

1. **Square integrability:** $\int_0^1 \varphi^2(u) du < \infty$
2. **Standardization:** Often $\int_0^1 \varphi(u) du = 0$ and $\int_0^1 \varphi^2(u) du = 1$
3. **Monotonicity:** Many score functions are monotone increasing
4. **Symmetry:** $\varphi(u) = -\varphi(1 - u)$ for signed-rank tests

Definition 2.2 (Score Generating Function). The score-generating function $\varphi(u)$ is standardized if:

$$\int_0^1 \varphi(u) du = 0, \quad \int_0^1 \varphi^2(u) du = 1$$

3 Hájek Representation Theorem

The Hájek representation (also known as the Hájek projection) is a fundamental result that expresses a linear rank statistic as a sum of independent random variables plus a remainder that becomes negligible as sample size increases.

Theorem 3.1 (Hájek Representation). Let $S_N = \sum_{i=1}^N c_{Ni} a_N(R_{Ni})$ be a linear rank statistic based on ranks $R_{N1}, \dots, R_{NN}$ of i.i.d. observations with continuous distribution. Under suitable conditions on the scores and constants, we have:

$$S_N = \mu_N + \sum_{i=1}^N c_{Ni} \varphi(F(X_i)) + o_p(\sqrt{N})$$

where μ_N is a centering constant, F is the true distribution function, and φ is the score function.

3.1 Intuition

The representation shows that a linear rank statistic behaves asymptotically like a sum of independent terms. This is crucial for establishing asymptotic normality.

3.2 Two-Sample Case

For the two-sample problem with $S = \sum_{i=1}^m a_N(R_i)$, the Hájek representation takes the form:

$$S - \mathbb{E}[S] = \frac{1}{m} \sum_{i=1}^m \varphi(U_i) - \frac{1}{n} \sum_{j=1}^n \varphi(V_j) + o_p(1)$$

where $U_i = F(X_i)$ and $V_j = G(Y_j)$.

Theorem 3.2 (Hájek Representation for Two-Sample Statistics). Let $S = \sum_{i=1}^m a_N(R_i)$ with scores generated by φ . Under H_0 (identical distributions):

$$\frac{S - \mathbb{E}[S]}{\sqrt{\text{Var}(S)}} = \frac{1}{\sqrt{N}} \sum_{i=1}^N \varphi\left(\frac{i}{N+1}\right) + o_p(1)$$

More precisely, there exists a projection $\hat{S}$ such that:

$$\|S - \hat{S}\|_2^2 = \mathbb{E}[(S - \hat{S})^2] = o(N)$$

and $\hat{S}$ is a sum of independent random variables.

3.3 The Projection

The Hájek projection $\hat{S}$ is given by:

$$\hat{S} = \mathbb{E}[S] + \sum_{i=1}^N (\mathbb{E}[S|X_i] - \mathbb{E}[S])$$

For linear rank statistics, this simplifies to:

$$\hat{S} = \mu_N + \sum_{i=1}^m \varphi(F(X_i)) - \sum_{j=1}^n \varphi(G(Y_j))$$

under appropriate standardization.

Example 3.1 (Hájek Representation for Wilcoxon). For the Wilcoxon rank-sum statistic with $a_N(i) = i$, the score function is $\varphi(u) = u$. Under H_0 , the Hájek representation gives:

$$\frac{W - \mathbb{E}[W]}{\sqrt{\text{Var}(W)}} \approx \frac{1}{\sqrt{mn/(m+n)}} \left(\frac{1}{m} \sum_{i=1}^m U_i - \frac{1}{n} \sum_{j=1}^n V_j \right)$$

where $U_i, V_j \stackrel{iid}{\sim} U(0, 1)$. This immediately suggests asymptotic normality.

4 Asymptotic Normality of Rank Tests

Using the Hájek representation, we can establish the asymptotic normality of linear rank statistics under general conditions.

4.1 Chernoff-Savage Theorem

The classical result on asymptotic normality of rank statistics was established by Chernoff and Savage (1958).

Theorem 4.1 (Chernoff-Savage). Consider the two-sample linear rank statistic $S = \sum_{i=1}^m a_N(R_i)$ with scores generated by φ satisfying:

- $\int_0^1 \varphi^2(u) du < \infty$
- φ is a difference of two monotone functions
- $m/N \rightarrow \lambda \in (0, 1)$ as $N \rightarrow \infty$

Then under H_0 (identical distributions):

$$\frac{S - \mathbb{E}[S]}{\sqrt{\text{Var}(S)}} \xrightarrow{d} N(0, 1)$$

4.2 Regularity Conditions

The typical conditions for asymptotic normality include:

1. **Score function:** φ is square-integrable and can be approximated by smooth functions
2. **Constants:** $\max_i c_{Ni}^2 / \sum c_{Ni}^2 \rightarrow 0$ (Noether condition)
3. **Sample sizes:** $m/N \rightarrow \lambda \in (0, 1)$
4. **Distribution:** Under H_0 , F is continuous

Theorem 4.2 (Asymptotic Normality). Under the Hájek representation and the above conditions:

$$\frac{S - \mathbb{E}[S]}{\sqrt{\text{Var}(S)}} \xrightarrow{d} N(0, 1)$$

where the variance can be estimated consistently by:

$$\text{Var}(S) = \frac{mn}{N(N-1)} \sum_{i=1}^N (a_N(i) - \bar{a}_N)^2$$

with $\bar{a}_N = \frac{1}{N} \sum_{i=1}^N a_N(i)$.

4.3 Variance Formulas

For common test statistics, we have explicit variance formulas:

- **Wilcoxon rank-sum:** $\text{Var}(W) = \frac{mn(N+1)}{12}$
- **General scores:** $\text{Var}(S) = \frac{mn}{N(N-1)} \sum_{i=1}^N (a_N(i) - \bar{a}_N)^2$
- **Van der Waerden:** $\text{Var}(S) \approx \frac{mn}{N-1}$ (for standardized normal scores)

4.4 Local Asymptotic Normality (LAN)

For studying power against local alternatives, we use Le Cam's LAN theory.

Definition 4.1 (Local Alternatives). Consider contiguous alternatives where the distributions differ by $O(1/\sqrt{N})$:

$$G_N(x) = F(x - \Delta/\sqrt{N})$$

for location shift, or more general parametric perturbations.

Under local alternatives, the rank statistic has asymptotic distribution:

$$\frac{S - \mathbb{E}_0[S]}{\sqrt{\text{Var}_0(S)}} \xrightarrow{d} N(\mu(\Delta), 1)$$

where $\mu(\Delta)$ depends on the score function and the alternative.

4.5 Pitman Efficiency

The asymptotic relative efficiency (ARE) of two rank tests is given by the ratio of their efficacy parameters:

$$\text{ARE}(T_1, T_2) = \left(\frac{\mu'_1(0)}{\sigma_1} \right)^2 / \left(\frac{\mu'_2(0)}{\sigma_2} \right)^2$$

For location shift with density f :

$$\text{ARE}(\text{Wilcoxon}, t) = 12\sigma^2 \left(\int f^2(x) dx \right)^2$$

which equals $3/\pi \approx 0.955$ for normal, and exceeds 1 for heavy-tailed distributions.

Example 4.1 (ARE for Common Distributions).

Distribution	Wilcoxon vs. t	Van der Waerden vs. t	Savage vs. t
Normal	0.955	1.0	0.5
Logistic	1.097	1.0	0.5
Laplace	1.5	1.0	0.75
Exponential	3.0	1.0	1.0
Cauchy	∞	∞	0

5 Applications to Common Tests

5.1 Wilcoxon Rank-Sum Test

For Wilcoxon, $\varphi(u) = u$ (or standardized version $\varphi(u) = \sqrt{12}(u - 1/2)$). The Hájek representation gives:

$$\frac{W - \mathbb{E}[W]}{\sqrt{\text{Var}(W)}} = \sqrt{\frac{12mn}{N(N+1)}} \left(\frac{1}{m} \sum_{i=1}^m U_i - \frac{1}{n} \sum_{j=1}^n U'_j \right) + o_p(1)$$

which is asymptotically $N(0, 1)$.

5.2 Van der Waerden Test

For normal scores, $\varphi(u) = \Phi^{-1}(u)$. This test is asymptotically equivalent to the optimal parametric test when the data are normal.

5.3 Savage Test

For exponential scores, $\varphi(u) = -\ln(1 - u) - 1$. This test is optimal for detecting scale differences in exponential distributions.

5.4 One-Sample Case

For one-sample signed-rank statistics:

$$S^+ = \sum_{i=1}^n \text{sign}(X_i) a_N(R_i^+)$$

where R_i^+ are ranks of $|X_i|$. The asymptotic theory is similar, with additional symmetry conditions.

Theorem 5.1 (Asymptotic Normality for Signed-Rank Statistics). Under H_0 (symmetric about 0) and appropriate conditions:

$$\frac{S^+}{\sqrt{\text{Var}(S^+)}} \xrightarrow{d} N(0, 1)$$

6 Technical Details and Extensions

6.1 Hájek's Projection Lemma

Lemma 6.1 (Hájek Projection). Let S be a statistic based on i.i.d. observations. The projection $\hat{S}$ that minimizes $\mathbb{E}[(S - \hat{S})^2]$ among all sums of independent terms is:

$$\hat{S} = \mathbb{E}[S] + \sum_{i=1}^N (\mathbb{E}[S|X_i] - \mathbb{E}[S])$$

6.2 Variance of the Projection

The variance of the projection satisfies:

$$\text{Var}(\hat{S}) = \sum_{i=1}^N \text{Var}(\mathbb{E}[S|X_i])$$

and $\mathbb{E}[(S - \hat{S})^2] = \text{Var}(S) - \text{Var}(\hat{S})$.

6.3 Contiguity and Local Power

For studying power, we use the concept of contiguity. If the sequence of alternatives is contiguous to the null, then the limiting distribution under the alternative can be obtained

via Le Cam's third lemma.

Theorem 6.1 (Le Cam's Third Lemma). If Λ_N is the log-likelihood ratio and under P_N :

$$\begin{pmatrix} T_N \\ \Lambda_N \end{pmatrix} \xrightarrow{d} N \left(\begin{pmatrix} 0 \\ -\sigma^2/2 \end{pmatrix}, \begin{pmatrix} \tau^2 & \rho \\ \rho & \sigma^2 \end{pmatrix} \right)$$

then under Q_N (the alternative):

$$T_N \xrightarrow{d} N(\rho, \tau^2)$$

This allows us to compute the asymptotic power of rank tests against local alternatives.

6.4 Score Function Optimality

For a given density f , the locally most powerful rank test uses score function:

$$\varphi(u) = -\frac{f'(F^{-1}(u))}{f(F^{-1}(u))}$$

This is the Fisher score function for location.

Example 6.1 (Optimal Score Functions). • **Normal:** $\varphi(u) = \Phi^{-1}(u)$ (Van der Waerden)

- **Logistic:** $\varphi(u) = u - 1/2$ (Wilcoxon)
- **Laplace:** $\varphi(u) = \text{sign}(u - 1/2)$ (median test)
- **Exponential:** $\varphi(u) = -\ln(1 - u) - 1$ (Savage)

7 Summary

Key Takeaways

- **Linear rank statistics:** $S = \sum c_i a_N(R_i)$ unify nonparametric tests
- **Score functions:** $\varphi : (0, 1) \rightarrow \mathbb{R}$ generate scores and determine test properties
- **Hájek representation:** Expresses rank statistic as sum of i.i.d. terms + negligible remainder
- **Asymptotic normality:** Follows from Hájek representation under mild conditions
- **Variance formulas:** $\text{Var}(S) = \frac{mn}{N(N-1)} \sum (a_N(i) - \bar{a}_N)^2$
- **Local alternatives:** Allow power calculation via Le Cam's theory
- **ARE:** Compares efficiency of different tests across distributions

8 Exercises

Exercise 8.1 (Score Functions). For each of the following tests, identify the score function $\varphi(u)$:

- (a) Mood's test for scale: $a_N(i) = \left(\frac{i}{N+1} - \frac{1}{2}\right)^2$
- (b) Klotz's normal scores test for scale: $a_N(i) = \left[\Phi^{-1}\left(\frac{i}{N+1}\right)\right]^2$
- (c) Ansari-Bradley test: $a_N(i) = \left|\frac{i}{N+1} - \frac{1}{2}\right|$

Exercise 8.2 (Hájek Representation). Derive the Hájek projection for the Wilcoxon signed-rank statistic $S^+ = \sum_{i=1}^n R_i^+ \mathbb{I}(X_i > 0)$ under H_0 that the distribution is symmetric about 0.

Exercise 8.3 (Asymptotic Normality). Show that the Mood test statistic for scale:

$$M = \sum_{i=1}^m \left(\frac{R_i}{N+1} - \frac{1}{2} \right)^2$$

is asymptotically normal under H_0 . Find its asymptotic mean and variance.

Exercise 8.4 (ARE Calculation). Compute the asymptotic relative efficiency of the Wilcoxon test to the t-test for the logistic distribution with density $f(x) = e^{-x}/(1+e^{-x})^2$.

Exercise 8.5 (Optimal Scores). For the Cauchy distribution with density $f(x) = 1/[\pi(1+x^2)]$, derive the optimal score function for location and show that it leads to a test with infinite ARE relative to the t-test.

Exercise 8.6 (Local Power). Consider the two-sample problem with $m = n = N/2$ and local alternative $\Delta/\sqrt{N}$. Derive the asymptotic power function for:

- (a) Wilcoxon test with $\varphi(u) = u - 1/2$
- (b) Van der Waerden test with $\varphi(u) = \Phi^{-1}(u)$

Assume the underlying distribution is normal with unit variance.

Exercise 8.7 (Projection Variance). For a general linear rank statistic $S = \sum_{i=1}^m a_N(R_i)$, show that:

$$\text{Var}(\hat{S}) = \frac{mn}{N(N-1)} \sum_{i=1}^N (a_N(i) - \bar{a}_N)^2$$

where $\hat{S}$ is the Hájek projection.

Exercise 8.8 (Conceptual Questions). Answer briefly:

- (a) Why is the Hájek representation important for asymptotic theory?
- (b) What conditions on score functions are needed for asymptotic normality?
- (c) Explain the concept of contiguity in the context of rank tests.
- (d) How does the choice of score function affect the power of a rank test?

- (e) Why are normal scores tests asymptotically efficient for normal data?

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